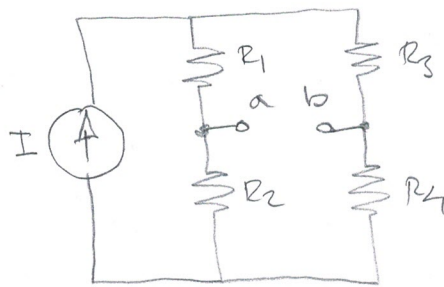


3.71)



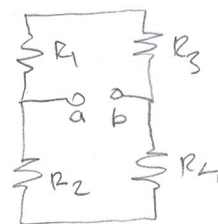
BESTÄM EN NORTON OCH EN
THEVENIN EK VÄRDET TILL
KRETSEN.

$$R_1 = 12 \Omega \quad R_3 = 5 \Omega$$

$$R_2 = 10 \Omega \quad R_4 = 2 \Omega$$

$$I = 3 \text{ A}$$

Lösning: Bestämmer först R_{TH} (som är samma som R_N)



RIKTA OM

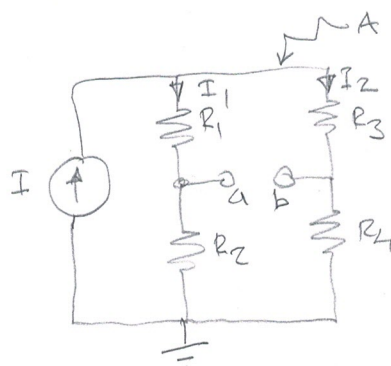
$\Rightarrow$



$$R_{ab} = \frac{(R_1 + R_3)(R_2 + R_4)}{R_1 + R_2 + R_3 + R_4} = 7.03 \Omega$$

$$R_{TH} = R_N$$

FÖRST THEVENIN SPÄNNINGEN.



KSTL GER

$$I - I_1 - I_2 = 0 \quad (1)$$

PAR

$$I_1 = \frac{V_A}{R_1 + R_2}, \quad I_2 = \frac{V_A}{R_3 + R_4} \quad (2)$$

(2) (1)

$$I = V_A \left(\frac{1}{R_1 + R_2} + \frac{1}{R_3 + R_4} \right) =$$

$$= V_A 0.18831 \text{ A}$$

$$V_A = \frac{3}{0.18831} \approx 15.93 \text{ VOLT}$$

SPÄNNINGEN V_{ab}

KSPL GER

$$R_2 I_1 - R_4 I_2 - V_{ab} = 0$$

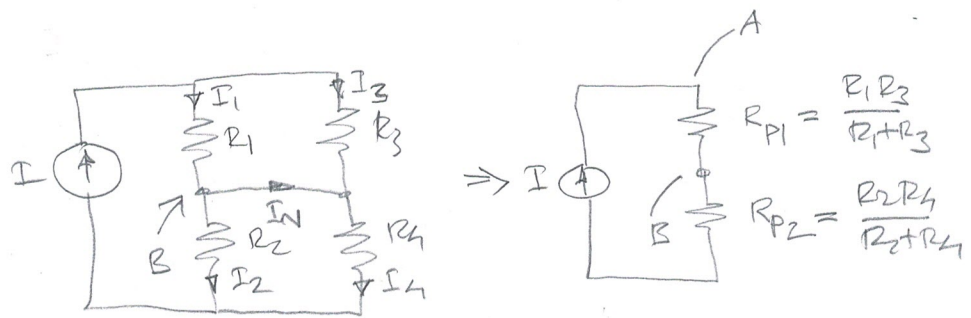
$$-V_{ab} = R_4 I_2 - R_2 I_1 = 2 \cdot \frac{15.93}{4} -$$

$$- 10 \cdot \frac{15.93}{22} = -2.68 \text{ VOLT}$$

AVVÄRT

$$V_{ab} = V_{TH} = 2.68 \text{ VOLT.}$$

NORBLANSSTRÖMMEN DÄ? DÄ HAR VI



SPÄNNINGEN I A $V_A = (R_{P1} + R_{P2}) I = 15.58 \text{ VOLT}$

$$V_B = R_{P2} I = 5 \text{ VOLT}$$

STRÖMMEN $I_1 = \frac{V_A - V_B}{R_1} = \frac{15.58 - 5}{12} = \underline{0.882 \text{ A}}$

KSTL FÖR NODEN B. (VÄNSTER)

$$I_1 - I_N - I_2 = 0$$

$$I_N = I_1 - I_2 = 0.882 - \frac{V_B}{R_2} =$$

$$= 0.882 - \frac{5}{10} = \underline{0.382 \text{ A}}$$

SVAR : $\begin{cases} R_{TH} = R_N = 7.03 \Omega \\ V_{TH} = 15.93 \text{ V} \\ I_N = 0.382 \text{ A} \end{cases}$